FINAL EXPLANATION: BIOLOGY GRADE 10

Student Assessment Document

FINAL EXPLANATION: BIOLOGY GRADE 10 — SUB-STRAND 2.2: PLANT TRANSPORT	
Student Assessment Document	
Student Name	____
Class	____
Date	____

INSTRUCTIONS FOR STUDENTS
This is your Final Explanation — your chance to show everything you have learned across all 8 lessons about plant transport. Your goal is to answer the driving question completely and confidently, using scientific evidence and the correct vocabulary you have built up. Read each section prompt carefully, then write your response in the space provided. Use the exemplars as a guide for the level of detail expected — but write in your own words. Remember: a strong Final Explanation always connects back to the farmer in Kibwezi. Her maize is your anchor. Explain the science so well that she would understand exactly why her crop is wilting — and what is happening inside every plant on her farm.

Section 1 — The Journey of Water: From Kibwezi Soil to the Highest Maize Leaf	
Describe the complete pathway that a water molecule takes from the soil outside a maize root all the way to the air above the highest leaf. In your answer: (a) name each tissue and structure the water passes through in order, (b) explain the driving force that pulls water upward at each stage, and (c) explain why the distance to the water table matters for the maize plants growing near the dry riverbed in Kibwezi.	A water molecule in the Kibwezi soil first enters the maize plant through root hair cells by osmosis. Root hair cells have a lower water potential than the surrounding soil solution (when soil moisture is adequate), so water moves down a water potential gradient into the cell. The water then moves through the root cortex — either through the cytoplasm of cells (symplast pathway) or through the cell walls (apoplast pathway) — until it reaches the endodermis. The Casparian strip in the endodermis blocks the apoplast route and forces all water into the symplast, acting as a checkpoint that controls which minerals enter the xylem. Water then passes into the xylem vessels in the root. Once inside the xylem, water is pulled upward through the stem and into the leaves by a process called transpiration pull (cohesion-tension mechanism). As water evaporates from the mesophyll cells inside the leaf through the stomata — a process called transpiration — it creates tension (negative pressure) in the xylem. Because water molecules are strongly attracted to each other (cohesion) and to the walls of the xylem vessels (adhesion), this tension pulls the entire water column upward continuously, like water rising through a very thin straw. This is called the cohesion-tension theory. Root pressure also contributes a small upward push, especially at night when transpiration is low. Finally, water reaches the leaf mesophyll cells and either stays to be used in photosynthesis or evaporates through the stomata into the atmosphere. The maize plants near the dry seasonal riverbed in Kibwezi wilt sooner because the water table there is much deeper. The xylem

	water column must span a greater height, and if the soil around the upper roots dries out, the water column can break (cavitation), and the cohesion-tension mechanism fails. There is simply not enough water close to the roots to maintain continuous flow, so the plant wilts even if some moisture remains deeper in the soil.

Section 2 — Transpiration: Why the Maize Loses Water Faster Than It Can Absorb It	
Explain what transpiration is and why it happens. Then use your understanding of stomata, guard cells, and environmental conditions to explain: (a) why the Kibwezi maize wilts specifically in the afternoon rather than in the morning, and (b) why the sukuma wiki growing in the shade of the mango tree barely wilts at all. Include at least TWO environmental factors in your answer.	Transpiration is the loss of water vapour from a plant, mainly through tiny pores on the leaf surface called stomata. It happens because the air spaces inside the leaf are humid (high water potential) while the outside air is drier (lower water potential), so water vapour moves down this gradient and diffuses out. Transpiration is not a mistake — it is an unavoidable consequence of leaves opening their stomata to let in carbon dioxide for photosynthesis. Stomata are opened and closed by pairs of guard cells. When guard cells absorb water by osmosis, they become turgid and bow outward, opening the pore. When they lose water, they become flaccid and the pore closes. Guard cells respond to light, carbon dioxide concentration, and water availability. The Kibwezi maize wilts in the afternoon — not the morning — because transpiration rate is highest in the afternoon. By midday and early afternoon, temperatures are at their peak (Kibwezi is semi-arid and can be very hot), which increases the kinetic energy of water molecules, speeding up evaporation. High temperature also reduces air humidity, steepening the water potential gradient between the leaf interior and the atmosphere. Wind speed may also be higher, carrying away humid air from around the leaf surface and further increasing water loss. All of these factors make the rate of transpiration far exceed the rate of water absorption by the roots, so the plant cells lose turgor and wilt. The sukuma wiki under the mango tree barely wilts because shade reduces two key drivers of transpiration: (1) Light intensity is lower, so guard cells receive less light stimulus and stomata remain more closed, reducing water vapour loss. (2) Temperature under the tree canopy is cooler, slowing evaporation. This means the sukuma wiki loses water much more slowly, and the rate of absorption through the roots keeps pace with the rate of loss — so the plant stays turgid and upright.

Section 3 — Translocation: Moving Food from Leaves to the Rest of the Plant	
The farmer in Kibwezi notices that even when her maize plants look healthy and green, the cobs (maize ears) sometimes fail to fill properly during a drought. Explain: (a) how the food (sugars) made in the leaves travels to the developing cob, naming the tissue and process involved, (b) the source-to-sink model, and (c) how severe water stress can disrupt translocation and affect grain filling in maize.	While water travels upward through xylem, the sugars produced by photosynthesis in the leaves travel through a different tissue called the phloem. The movement of dissolved sugars (mainly sucrose) and other organic compounds through the phloem is called translocation. Unlike xylem transport, which is passive and driven by physical forces, translocation requires energy (ATP) at key loading and unloading steps. The process is explained by the pressure-flow (mass flow) hypothesis. Leaves are called the source — they actively load sucrose into the phloem sieve tubes using companion cells and ATP-powered carrier proteins. This increases the solute concentration inside the phloem at the source, so water enters the phloem by osmosis from nearby xylem, creating high hydrostatic pressure. At the other end, the developing maize cob (and roots) are called sinks — they unload sucrose for growth and energy, reducing solute concentration. Water leaves the phloem at the sink by osmosis, lowering pressure. The pressure difference between source and sink drives a continuous mass flow of phloem sap from source to sink. During severe drought in Kibwezi, water stress disrupts translocation in two ways. First, if leaves lose too much water,

	photosynthesis slows or stops — so less sugar is produced at the source, reducing the pressure gradient. Second, the companion cells that actively load sucrose into the phloem require ATP, and a severely stressed plant redirects resources away from growth. The result is that the developing maize cob (sink) receives insufficient sucrose for grain filling, producing small, poorly developed cobs — even on plants that appeared outwardly healthy earlier in the season. This is a hidden cost of drought that the farmer may not connect directly to water stress.

Section 4 — Why Some Plants Recover and Others Die: Turgor, Permanent Wilting, and Adaptation	
The farmer observes that by evening, some wilted maize plants recover, while others do not. Using your knowledge of turgor pressure, the permanent wilting point, and plant adaptations, explain: (a) the difference between temporary wilting and permanent wilting, (b) the cellular reason why a plant that passes the permanent wilting point cannot recover, and (c) at least TWO structural or behavioural adaptations that help plants in semi-arid areas like Kibwezi reduce water loss.	Plant cells stay firm and upright because of turgor pressure — the pressure of water pushing outward against the cell wall when a cell is full of water (turgid). When a plant loses water faster than it absorbs it, cells lose turgor and become flaccid, causing the plant to wilt. This is called temporary wilting if the soil still contains water above the permanent wilting point. In temporary wilting, as temperatures drop in the evening, transpiration slows and the plant can reabsorb enough water from the soil to regain turgidity — which is why some of the Kibwezi maize recovers overnight. However, if the soil water potential drops so low that the roots can no longer absorb water by osmosis — even with stomata fully closed — the plant has reached its permanent wilting point. At this stage, cells have been plasmolysed (the cell membrane has pulled away from the cell wall due to excessive water loss). The normal osmotic gradient needed to pull water in no longer exists, and cellular machinery begins to break down. Even if you water the soil at this point, the damage to membranes, enzymes, and transport proteins may be irreversible. This is why some of the farmer's maize plants do not recover — they passed the permanent wilting point during the hottest part of the afternoon. Plants adapted to semi-arid conditions like Kibwezi have evolved several features to delay reaching this point: (1) Sunken stomata — some drought-adapted plants have stomata in pits recessed below the leaf surface, which traps humid air and reduces the water potential gradient, slowing transpiration. (2) Thick, waxy cuticle — a thick waxy layer on the leaf surface (as seen in succulent plants) prevents water from evaporating through the leaf surface directly, reducing cuticular transpiration. (3) Stomatal closure during heat — CAM plants open stomata only at night, fixing CO_2 when temperatures are cool and transpiration losses are minimal. (4) Deep or wide root systems — roots that spread broadly or reach deeper soil layers access more water, maintaining the water column even during dry spells.

Section 5 — Answering the Driving Question: A Complete Explanation for the Kibwezi Farmer	
You are writing a letter to the maize farmer in Kibwezi. Using everything you have learned across all 8 lessons, write a full scientific explanation that answers the driving question: How does water and food travel through an entire plant — from the soil to the highest leaf — and why does this journey sometimes fail, causing crops like maize in Kibwezi to wilt and	Dear Farmer, Thank you for sharing your observations — they reveal something fascinating about how plants work, and understanding the science behind what you have seen can help you protect your harvest. Inside every maize plant on your farm, there is an incredible transport system made of two types of tubes running from roots to leaves. The first is the xylem, which carries water and dissolved mineral salts upward from the soil. Water enters through the root hair cells by osmosis — moving from the wet soil into the root because the root cells have a lower water potential. It then travels up through the xylem vessels in the stem, pulled upward by a force called transpiration pull. As water evaporates from the leaves through tiny pores called stomata, it creates a tension that draws more water up from the roots — a bit like drinking through a long straw. The second transport tissue is the phloem, which carries the sugars made by photosynthesis in the leaves downward (and

die? Your explanation must: (a) describe both xylem (water) and phloem (food) transport, (b) explain the specific reasons her maize wilts in the afternoon, (c) explain why some plants recover and others die, (d) explain why the shaded sukuma wiki survives better, and (e) suggest ONE practical action she could take, based on the science, to protect her maize during the dry season.	upward) to wherever the plant needs energy — including the developing maize cob. This process is called translocation and is driven by pressure differences between sugar-producing leaves (sources) and sugar-using organs like the cob (sinks). Now, why does your maize wilt in the afternoon? By midday in Kibwezi, the temperature is very high and the air is dry. These conditions cause your maize leaves to lose water through the stomata far faster than the roots can absorb it from the soil. Even if you watered the soil that morning, the plant simply cannot move water upward quickly enough to replace what is being lost. The cells lose their water pressure (turgor) and the plant droops. This is not a failure of your watering — it is the physics of water movement struggling to keep up with the heat of a semi-arid afternoon. The plants near the dry riverbed wilt sooner because the soil there dries out faster and the water table is deeper. The xylem water column breaks more easily with less water in the soil, stopping transport altogether. Why do some plants recover in the evening but others do not? As the sun sets and temperatures fall, transpiration slows. Plants that still have enough soil water can reabsorb it and regain their turgor — they were only temporarily wilted. But plants that lost too much water reach what scientists call the permanent wilting point: the soil is so dry that the roots cannot pull water in even when the stomata are closed. These plants suffer cell damage and cannot recover, which is why some of your maize does not revive overnight. Your neighbour's sukuma wiki survives better under the mango tree because shade reduces two things: light intensity (so the stomata open less and lose less water) and temperature (so evaporation is slower). The rate of water loss matches the rate of water supply, and the plants stay turgid all day. One practical step you could take based on this science: apply a layer of mulch (such as dry maize stalks or grass) around the base of your maize plants. Mulch reduces soil surface temperature, slows evaporation from the soil, and helps the soil retain the water you apply in the morning for longer. This keeps water available to the roots during the hottest afternoon hours, helping to delay wilting — and giving more of your plants a chance to recover each evening. Your observations were not just puzzling — they were science. I hope this explanation helps. Yours in biology, [Student Name]

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Criterion	**Excellent (4)**	**Proficient (3)**	**Developing (1–2)**
Scientific Accuracy of Transport Mechanisms	Accurately and fully explains both xylem (cohesion-tension, osmosis, root pressure, Casparian strip) and phloem (pressure-flow/mass flow, source-sink model, active loading) transport using correct terminology throughout. No	Correctly explains both xylem and phloem transport with mostly accurate terminology. Minor omissions (e.g., Casparian strip or active loading not mentioned) but no significant misconceptions.	Attempts to describe transport in xylem and/or phloem but contains one or more misconceptions (e.g., confuses xylem and phloem, states that plants 'suck up' water without explaining the mechanism, or omits the role of osmosis at the root).

	misconceptions present.		
Explanation of Wilting, Recovery, and Permanent Wilting Point	Clearly distinguishes temporary from permanent wilting with reference to turgor pressure, plasmolysis, and the permanent wilting point. Accurately explains at the cellular level why some Kibwezi maize recovers in the evening and others do not, linking soil water potential to root absorption capacity.	Correctly explains the difference between temporary and permanent wilting and connects it to the Kibwezi scenario. Cellular-level explanation (e.g., plasmolysis) may be incomplete but the overall reasoning is sound.	Recognises that some plants wilt permanently but cannot explain why at the cellular or tissue level. May simply state 'the plant ran out of water' without reference to turgor pressure, water potential, or the permanent wilting point.
Application to the Anchoring Phenomenon (Kibwezi Maize)	Every section is explicitly and meaningfully connected to the Kibwezi farmer's observations (afternoon wilting, riverbed vs. slope location, shaded sukuma wiki). Scientific concepts are used to explain each specific observation, not just described in general terms.	Most sections connect back to the Kibwezi phenomenon with reasonable specificity. At least three of the farmer's specific observations (afternoon timing, location difference, shade effect) are explained using scientific reasoning.	Some reference to Kibwezi is made but the connections are superficial or generic (e.g., 'the plant wilts because it is hot in Kibwezi') without applying the underlying transport mechanisms to explain the specific observations.
Use of Scientific Vocabulary	Consistently and correctly uses all key terms: xylem, phloem, transpiration, translocation, osmosis, water potential, turgor pressure, cohesion-tension, stomata, guard cells, source, sink, permanent wilting point, plasmolysis, and Casparian strip. Terms are used in context, not just listed.	Uses most key terms correctly in context. May omit 2–3 terms (e.g., plasmolysis or Casparian strip) but demonstrates clear understanding of the core vocabulary: xylem, phloem, transpiration, translocation, osmosis, turgor pressure, stomata, and guard cells.	Uses fewer than half of the expected key terms, or uses several terms incorrectly. Explanations rely heavily on everyday language (e.g., 'tubes carry water up,' 'leaves breathe') rather than scientific vocabulary.
Quality of Scientific Reasoning and Evidence-Based Argument	Explanations follow a clear cause-and-effect chain throughout. Claims are supported by mechanisms (not just observations). The final section (letter to farmer) synthesises all concepts into a coherent, complete answer to the driving question and includes a scientifically justified practical recommendation.	Reasoning is mostly logical and cause-and-effect relationships are evident in most sections. The final letter addresses the driving question adequately and includes a practical recommendation, though the justification may lack full scientific depth.	Explanations are descriptive rather than mechanistic — the student describes what happens but not why or how. The final letter may restate observations without providing a scientific explanation, or the practical recommendation is absent or not grounded in the science learned.